

LEONAL ROBIN

Dindigul, Tamil Nadu, India

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SUMMARY

Computer Science and Engineering graduate with strong expertise in scalable backend services, distributed systems, and AI/ML. Proven ability to design, develop, and optimize robust software solutions, leveraging data structures, algorithms, and containerization for high-performance applications. National-level Hackathon Winner with a track record of innovative, collaborative engineering.

SKILLS SUMMARY

Programming Languages: Python, Java, Go, JavaScript, TypeScript, SQL

Frameworks & Libraries: React.js, Next.js, Node.js, Django, Flask, FastAPI, Spring Boot, RAG, LLM, Langchain, Databricks

Tools & Technologies: Docker, Kubernetes, MySQL, PostgreSQL, MongoDB, AWS, Git, Jira, Pinecone, FAISS, Firebase, Tailwind CSS

EXPERIENCE

Flam | Front-End Engineer (R&D) Intern

Bangalore, karnataka, India

NextJs, TypeScript, TailwindCSS, GitHub | Aug 2025 -- Mar 2026

- Engineered a Mastra + OpenAI function-calling pipeline for prompt-driven AI control, enabling image-to-video generation and dynamic canvas state management via NLP commands.
- Engineered a grouped QR scanning system to bundle and manage multiple related AR experiences, optimizing content delivery and user access.
- Developed a full-featured AR experience dashboard, implementing robust asset management, content organization, and publishing workflows for image and ground-tracking experiences.

MANNIT | SDE Intern

Chennai, Tamil nadu, India

ReactJs, TypeScript, LLMs, VectorDB, Flask, Langchain, FastAPI | Feb 2025 -- Jul 2025

- Architected an admin-driven vector database pipeline with automatic category generation and dynamic user-specific data isolation, ensuring real-time scalability with data growth.
- Engineered a hybrid FAISS local indexing layer with in-memory storage to resolve Pinecone API quota limits, achieving 3-5x faster similarity search and reducing query latency to milliseconds.
- Developed PostgreSQL-backed authentication, chat history persistence, and an admin dashboard for efficient embedding management.
- Developed an OAuth-authenticated GitHub app for AI-powered code reviews, providing quality improvements, architectural explanations, refactoring suggestions, and auto-generated Mermaid flowcharts.
- Engineered scalable codebase analysis for 10,000+ file repositories via AST parsing, smart filtering, and batched asynchronous processing.

Telesto Energy | SDE Intern

Coimbatore, Tamil Nadu, India

Django, SQL, Node.js, PlotlyJS, MySQL, Jira, REST | Feb 2025 -- Jul 2025

- Developed and optimized Node.js REST APIs to ingest and serve large-scale, real-time energy datasets with consistently low latency.
- Normalized and optimized MySQL schemas to enhance query performance and ensure data integrity across critical energy monitoring pipelines.
- Integrated ML models into the data pipeline for predictive analytics on energy consumption, automating inference on data streams and surfacing insights via PlotlyJS dashboards.
- Built a responsive Django web interface with real-time PlotlyJS visualizations, enabling users to monitor live energy metrics, analyze historical trends, and make data-driven decisions.

PROJECTS

Fake News Detection blue Certificate blue Demo

Nov 2024 -- Jan 2025

ReactJs, TailwindCSS, Flask, GitHub, AWS, LLM

- Built a full-stack task management platform with a React/TypeScript frontend and Spring Boot REST API backend for comprehensive task lifecycle management.
- Designed RESTful APIs integrated with MongoDB to persist task details, owners, shell commands, and execution history with timestamped output tracking.
- Implemented Kubernetes-based command execution by dynamically creating pods to run task commands in isolated container environments.
- Incorporated Chandy-Lamport snapshot and ring election concepts for distributed fault tolerance, modeling concurrent behavior with goroutines and channels.
- Simulated failures, recovery behavior, timeout handling, commit decisions, and abort scenarios to validate system resilience and fault tolerance.
- Built a Go-based simulation of disaster recovery in distributed transaction processing systems, implementing coordinator-participant flow with two-phase commit.
- Disaster Recovery System (May 2024 -- Jun 2024) - Go, Goroutines, Channels, Two-Phase Commit, Distributed Systems

- Automated build and deployment using GitHub Actions, Docker, Docker Hub, KIND, and Kubernetes manifests for backend, frontend, and MongoDB services.

EDUCATION

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering | Sep 2021 -- May 2025

Coimbatore, India

ACHIEVEMENTS AND ACTIVITIES

- NASA Space App Challenge (Oct 2024) — Seismic Data from Apollo and Mars InSight Missions
- ANSAN -- GDSC Project (Jan 2024) — Google Solutions Challenge India Regional Bootcamp
- Pragyan '23 -- National Winner (Mar 2023) — National Level Hackathon at NIT Trichy